A Game-Theoretic Model for Multi-Agent Reference-Conditioned Forgery Detection

Course project proposal — Game Theory

Abstract

We propose a detection system in which three modality-specialised reinforcement learning agents examine a *reference* image (the authoritative pattern) and a *test* image (the candidate), emit embeddings and belief reports, and a separate judge agent issues a genuine/forged verdict. We formalise the system as a finite-horizon partially observable stochastic game, show that the naive formulation is a degenerate (identical-interest) game with no strategic content, and propose two modifications that give it genuine game-theoretic structure: an adversarial forger that turns the outer problem into a two-player zero-sum Stackelberg game, and a market scoring rule that makes truthful reporting individually optimal for each expert. We specify the reward functions, the judge architecture, and a three-stage optimisation procedure (supervised pretraining, CTDE multi-agent PPO, PSRO population play), together with evaluation metrics that include exploitability and Shapley-value attribution.

1 Problem setup

The system receives a pair (x_R, x_T) and must decide whether x_T conforms to the pattern exhibited by x_R. The reference is treated as ground truth: the question is never "is x_T a plausible image" but "is x_T consistent with x_R". This is a paired-verification problem — signature verification, document authentication, seal or watermark checking, product-label inspection — rather than a standalone classification problem. Three experts extract complementary evidence:

- **Visual expert** — texture statistics, edge coherence, blending artefacts, resampling traces.
- **Text expert** — glyph shapes, kerning, baseline drift, layout geometry, OCR consistency against the reference.
- **Signal expert** — frequency-domain residuals, sensor noise pattern (PRNU-like), compression fingerprints.

Each expert produces an embedding pair (e_i^R, e_i^T) and a belief report. A judge consumes only the reports — never the raw pixels — and decides.

2 Formal model

Because the label is hidden, the agents observe different slices of the input, and the decision unfolds over several steps, the correct formalism is a finite-horizon *partially observable stochastic game*, i.e. a Dec-POMDP when payoffs are shared:

$$\mathcal{M} \;=\; \big\langle \mathcal{N},\; \mathcal{S},\; \{\mathcal{A}_i\},\; P,\; \{\mathcal{O}_i\},\; \{R_i\},\; \gamma,\; H \big\rangle.$$

Agents. $\mathcal{N} = \{1, 2, 3, J\}$, where $1, 2, 3$ are the visual, text and signal experts and J is the judge. In the adversarial extension (Section 3) a fifth agent F, the forger, is added.

State.

$$s_t \;=\; \big(x_R,\; x_T,\; y,\; m_t,\; b_t,\; t\big),$$

where $y \in \{0, 1\}$ is the hidden label (1 = forged), $m_t = \{(e_i^R, e_i^T, q_i)\}$ is the set of reports emitted so far, $b_t = \Pr(y = 1 \mid h_t)$ is the judge's running belief, and t is the step index. Episodes begin by sampling $(x_R, x_T, y) \sim \mathcal{D}$.

Observations. $o_i = \phi_i(x_R, x_T)$ is expert i's modality slice; $o_J = m_t$. The judge's exclusion from the raw input is the load-bearing assumption of the design. If the judge can see the images, the experts become decorative and no communication protocol needs to emerge.

Actions. Two levels, and the distinction matters for tractability.

- Expert i at step t takes $a_{i,t} = (\ell_{i,t},\, q_{i,t})$, where $\ell_{i,t}$ is a low-dimensional *control* action (which region, scale or frequency band to attend to; or STOP) and $q_{i,t} \in \Delta(\{0, 1\})$ is a scalar belief report.
- The judge takes $a_J \in \{\text{DECLARE-GENUINE},\ \text{DECLARE-FORGED},\ \text{QUERY-}i\}$, where each query costs c.

Design warning. The high-dimensional embedding must *not* be the RL action. Policy-gradient variance in a 256-dimensional continuous action space is prohibitive. Embeddings are produced by differentiable encoders trained by backpropagation; only the control decisions, the confidence report and the stopping decision are learned by RL.

Transition and horizon. Transitions are deterministic given the initial draw: $m_{t+1} = m_t \cup \{a_{i,t}\}$ and $b_{t+1} = \text{update}(b_t, a_{i,t})$. The episode terminates when the judge declares or when $t = H$. With $K = 3$ expert steps plus one decision, $H = 4$. A discount $\gamma < 1$ can be used to reward early stopping, though the explicit query cost c is the cleaner instrument.

3 Is this a game?

As stated, technically yes; substantively no. With a single shared reward the system is a *common-payoff* (identical-interest) stochastic game. This has one attractive formal property: identical payoffs make it an exact potential game whose potential function is the shared return, so best-response dynamics converge to a local equilibrium. But there is no strategic tension anywhere in it. No agent ever wants something another agent does not want, and every Nash equilibrium is simply a local optimum of a single scalar objective. The word "game" could be deleted without loss; what remains is distributed optimisation.

To obtain genuine game-theoretic content we adopt two modifications.

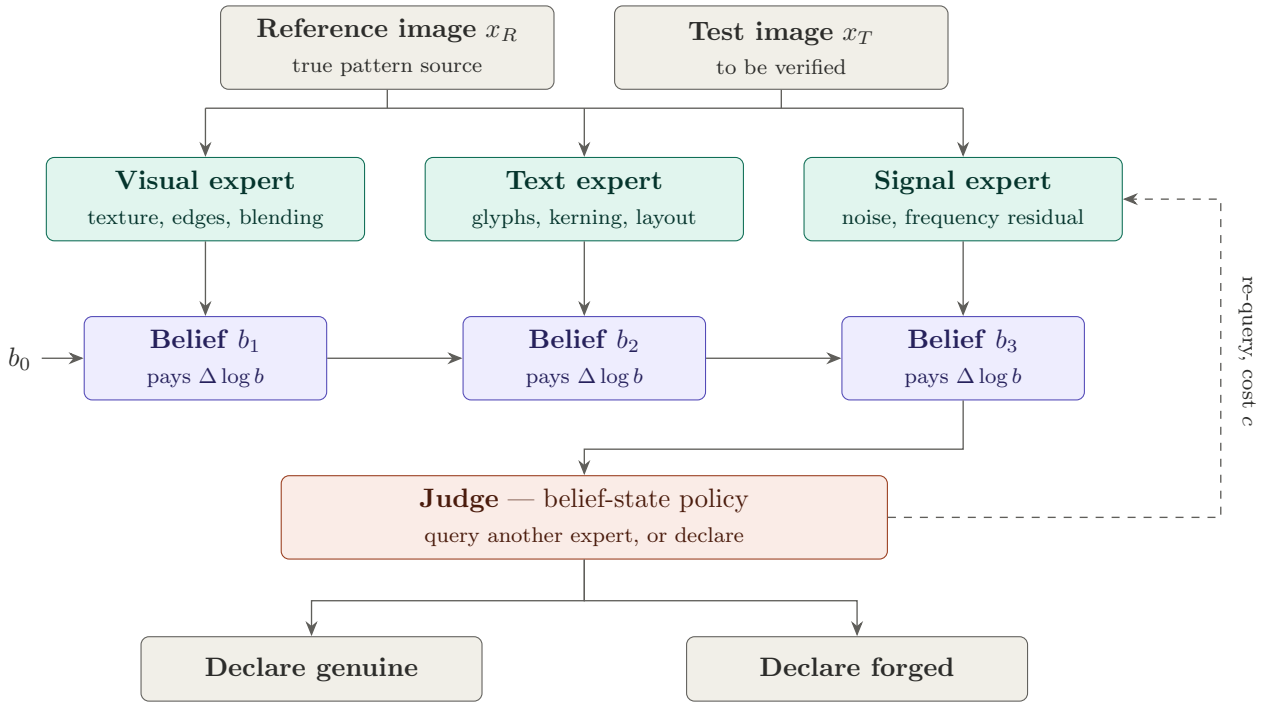


Figure 1: One detection episode (the inner game). The reference and test images are fed to three modality-specialised experts, each of which sees only its own slice of the input. Experts report sequentially into a shared belief b_t, initialised at the prior b_0; under the market scoring rule each expert is paid the log-difference it induces, $r_t = \log b_t(y) - \log b_{t-1}(y)$, so truthful reporting is myopically optimal and the payments telescope to the team objective. The judge observes only the reports — never the pixels — and at each step chooses between paying c to query a further expert (dashed return path) and declaring a verdict. Stopping after b_1 or b_2 is permitted and is the mechanism through which the query cost shapes behaviour.

3.1 Outer game: a two-player zero-sum Stackelberg game

Introduce a forger F that generates or perturbs test images, and let the detection team $D = \{1, 2, 3, J\}$ classify. Payoffs are

$$u_D = \mathbb{E}\big[\text{score}\big], \qquad u_F = -\,u_D,$$

so the interaction is zero-sum. Over mixed strategies the minimax theorem applies and the equilibrium detector is exactly the minimax-robust one. Because the detector is deployed first and the attacker adapts afterwards, the realistic threat model is a *Stackelberg* game with the detector as leader and the forger as follower. This also connects the project to the classical statistical framing: Wald's treatment of hypothesis testing as a zero-sum game against Nature, in which the minimax test corresponds to the least favourable prior. Our system is a learned approximation to that minimax test.

3.2 Inner game: information aggregation with incentives

Within the detection team, three further structures are available, of which we adopt the first two:

Sender–receiver game. Experts hold private evidence and send messages to a judge who acts. Even with perfectly aligned preferences this game has an uninformative "babbling" equilibrium

alongside the informative one; the equilibrium-selection problem is observable empirically as runs in which the embeddings collapse to a constant. Giving one expert a biased objective recovers the Crawford–Sobel partition-equilibrium structure.

Coalitional game. Define $v(S)$ as the expected score the judge achieves using only the experts in $S \subseteq \{1, 2, 3\}$. With three experts there are only eight coalitions, so *exact* Shapley values are computable — no Monte Carlo approximation is needed. Training the judge with random expert dropout makes $v(S)$ well defined.

Public-goods game (optional). Charging each expert for its own compute, $u_i = \alpha G - c_i(\text{effort}_i)$, produces a free-rider problem in which Nash effort is strictly below the social optimum. We note this as an extension rather than adopting it, to keep the project scoped.

4 Reward design

4.1 Judge: a strictly proper scoring rule

The judge reports $q_J \in [0, 1]$ and receives

$$r_J \;=\; \log q_J(y) \;-\; c \cdot \#\{\text{queries}\} \;-\; \big[\, c_{\text{FP}} \mathbb{1}\{\hat{y} = 1, y = 0\} + c_{\text{FN}} \mathbb{1}\{\hat{y} = 0, y = 1\} \,\big].$$

Properness means the payoff-maximising report *is* the judge's true posterior: the reward is incentive-compatible by construction, which is a mechanism-design argument rather than an engineering convenience. Plain 0/1 accuracy has no such property and should not be used as the training signal. In verification settings $c_{\text{FN}} > c_{\text{FP}}$ is usually appropriate — missing a forgery costs more than a false alarm.

4.2 Experts: a market scoring rule

Experts report sequentially starting from the prior b_0. The expert acting at step t receives

$$r_t \;=\; \log b_t(y) \;-\; \log b_{t-1}(y).$$

Two consequences follow. First, the payments telescope,

$$\sum_{t=1}^{K} r_t \;=\; \log b_K(y) - \log b_0(y),$$

so the scheme is *exactly budget-balanced* against the team objective: no credit is invented or destroyed. Second, each expert maximises its own payment by moving the running belief to its true posterior given its private evidence, so truthful reporting is myopically optimal. The non-myopic case — an expert withholding evidence early in order to profit from a larger correction later — is a known bluffing phenomenon in prediction markets and is worth reporting as an observed failure mode if it appears.

4.3 Alternative credit assignment

Where coalitional attribution is preferred, use the exact Shapley value

$$\phi_i \;=\; \sum_{S \subseteq \mathcal{N} \setminus \{i\}} \frac{|S|!\,(n - |S| - 1)!}{n!} \big[\, v(S \cup \{i\}) - v(S) \,\big],$$

computable in closed form at $n = 3$, or the cheaper difference reward $D_i = G(\mathbf{z}) - G(\mathbf{z}_{-i}, \bar{z}_i)$, which replaces expert i's output with a default (mean) embedding.

4.4 Shaping

All auxiliary terms are added as potential-based shaping, $F = \gamma\Phi(s') - \Phi(s)$, which provably leaves the optimal policy unchanged:

- $\Phi = -$ negative log loss of the current belief (progress toward certainty);
- contrastive alignment: $\|e_i^R - e_i^T\|$ small for genuine pairs, large for forged;
- a decorrelation penalty across the three experts — without it they collapse onto the same features and the coalitional analysis becomes vacuous.

5 The judge

The judge is the component most easily got wrong, so we specify it explicitly.

Capacity is deliberately limited. If the judge is as expressive as the experts it learns to ignore their messages and solve the task from whatever signal leaks through; the information asymmetry is the entire mechanism. We restrict the judge to a two-layer MLP or a two-layer transformer encoder and never give it raw pixels. This mirrors the prover–verifier setting, in which a trusted but weaker verifier tries to choose the correct answer while a more powerful untrusted prover tries to persuade it, and a justification protocol is expected to emerge from the game.

Architecture options. Table 1 summarises four choices in increasing order of theoretical yield. We adopt (d).

Design	Mechanism	Trade-off
(a) Set encoder	DeepSets or small transformer with a learned `[CLS]` token over message tokens	Permutation-equivariant, variable report count, works directly with expert dropout
(b) Recurrent updater	GRU over the message sequence; hidden state *is* the belief	Natural fit for the sequential MSR; easy to read b_t at every step
(c) Log-odds accumulator	$\lambda_t = \lambda_{t-1} + \Delta_i$, $b_t = \sigma(\lambda_t)$; judge is nearly parameter-free	Fully interpretable per-expert contribution, but forecloses cross-modal interactions
(d) Learned LLR + SPRT	Experts emit calibrated log-likelihood ratios; judge stops when λ_t exits (a_0, a_1)	The stopping rule becomes an optimisation problem with a known optimum rather than a heuristic

Table 1: Judge architectures. Option (d) makes the query-cost term principled: the SPRT is Bayes-optimal in minimising the expected number of samples for a given error bound. The complication is that our horizon is finite ($K = 3$), where the optimal rule formally requires backward induction; recent work learns that backward induction from data instead.

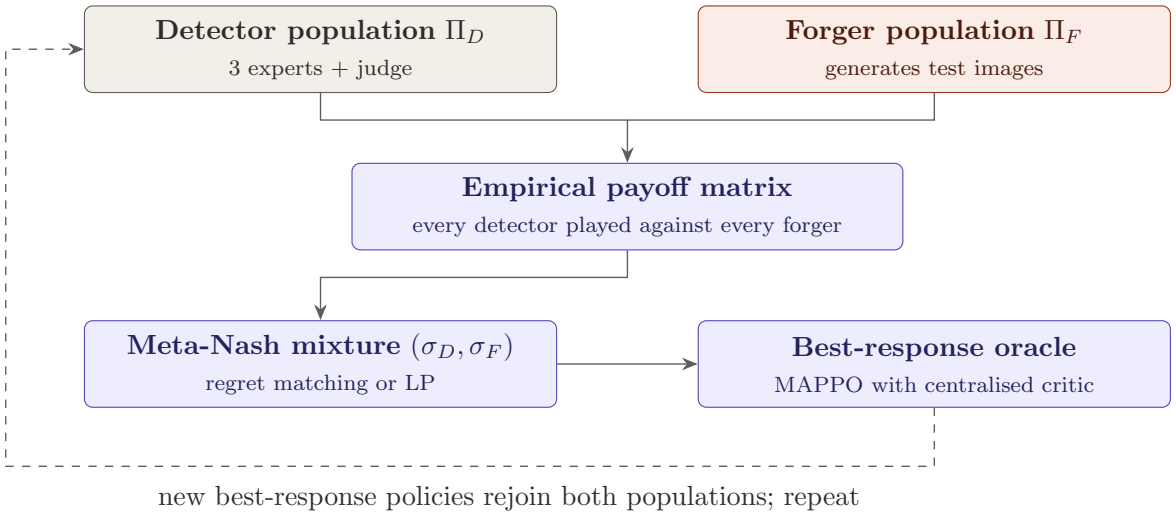


Figure 2: The outer adversarial loop (PSRO). Each detector policy in Π_D is played against each forger policy in Π_F — every cell of the payoff matrix is one full run of Figure 1 over a validation batch. A meta-solver computes the equilibrium mixture over the current populations, an approximate best response is trained against that mixture with MAPPO, and the new policy is added to its population. Because the meta-game is an explicit normal-form game that can be printed and inspected, this loop is what makes the project a game-theory artefact rather than a deep-learning one; it also yields exploitability (NashConv) directly as a robustness metric.

Heads. Whichever backbone is chosen, the judge carries three outputs: a probability head trained with the proper scoring rule, a query-policy head (actor over $\{\text{QUERY-}i, \text{DECLARE}\}$), and a value head for the centralised critic. Only the probability head requires calibration; we report expected calibration error and a reliability diagram, not accuracy alone.

Training schedule. Alternate rather than co-train. Freeze the experts and fit the judge by supervised learning on their current messages; then freeze the judge and improve the experts by RL against it. This is the checkability-training pattern, and it exists because simultaneous training tends to collapse into a private code between experts and judge that does not survive the judge being replaced.

Diagnostic. Swap in a freshly initialised judge and measure surviving accuracy. If accuracy collapses, the experts learned a private protocol rather than genuinely informative messages, and the Shapley attribution is measuring nothing.

6 Optimisation

Stage 1 — supervised pretraining of encoders. Metric learning (triplet loss or InfoNCE) on (x_R, x_T) pairs. Representations are *not* learned from scratch by RL; the sample complexity of doing so is prohibitive.

Stage 2 — cooperative MARL under CTDE. Centralised training with decentralised execution, using **MAPPO** with a centralised critic. This is the right default here: after the action-space

reduction of Section 2 the action spaces are small, there are only four agents, and on-policy PPO is markedly more stable than MADDPG once the environment becomes non-stationary through the addition of the forger. QMIX is a poor fit — its monotonic value factorisation imposes a structural assumption that buys nothing at $n = 3$. Credit assignment uses the market scoring rule of Section 4.2, with Shapley or difference rewards as comparison baselines.

Stage 3 — the adversarial outer loop. This is a nonconvex–nonconcave saddle-point problem; naive simultaneous gradient descent–ascent cycles rather than converges. Two defensible options:

- **PSRO / double oracle** (Figure 2). Maintain populations of detector and forger policies, construct the empirical payoff matrix, solve the meta-game with a Nash or regret-matching meta-solver, add a best response, repeat. This is the preferred choice because the meta-game is an explicit normal-form game that can be presented and analysed directly.
- **Optimistic gradient descent–ascent** (or extragradient), a cheaper single-loop method with last-iterate convergence guarantees in the monotone case.

7 Evaluation

We report both detection metrics and game-theoretic metrics:

- **Detection:** accuracy, AUC, equal error rate, expected calibration error, and the speed–accuracy curve (accuracy against mean number of experts queried).
- **Game-theoretic:** exploitability (NashConv) against a best-response attacker trained from scratch; exact Shapley values ϕ_1, ϕ_2, ϕ_3 as contribution attribution requiring no ablation retraining; and whether the Shapley allocation lies in the core of v.
- **Protocol robustness:** accuracy retained under judge replacement (Section 5), and under expert dropout.

8 Related work

Closest architecture. Recent multi-agent forensic reasoning frameworks for deepfake video detection use specialised agents examining texture, lighting, motion and physics, with a judge agent aggregating their findings into a final verdict — structurally very close to our pipeline, but treating the judge as a heuristic aggregator with no incentive analysis. *DeepForgeSeal* applies multi-agent adversarial reinforcement learning to watermark-based deepfake detection.

Judge–expert games. Irving, Christiano and Amodei's *AI Safety via Debate* introduces two agents competing in a zero-sum game to convince a judge of the truth or falsity of a claim, where the judge evaluates competing arguments rather than solving the problem. Anil et al.'s *Prover–Verifier Games* is the more rigorous treatment, analysing simultaneous and sequential variants and isolating the subset of games that provably have the desired equilibria.

Optimisation. PSRO generalises double oracle, iterated best response and fictitious play by computing approximate best responses to mixtures of policies with deep RL and applying empirical

game-theoretic analysis to select meta-strategies. MAPPO establishes that PPO-based multi-agent methods match or exceed competitive off-policy baselines in both final return and sample efficiency.

Credit assignment and mechanism. Shapley Q-value (SQDDPG) argues that a shared global reward gives each agent an inaccurate signal about its own contribution; SHAQ and Shapley Counterfactual Credits extend the theory. Hanson's market scoring rule — a sequentially shared proper scoring rule in which forecasters are scored by how much they improve or worsen the running report — supplies our expert payment scheme.

Claimed gap. The multi-agent deepfake literature has the architecture but no incentive analysis; the debate and prover–verifier literature has the game theory but operates on text rather than paired reference–test verification. This proposal occupies the middle: an explicit market scoring rule for expert payments, exact Shapley values for attribution, and a zero-sum outer game whose solution is a learned approximation to the minimax hypothesis test.

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